

CHEMISTRY & BIOCHEMISTRY

Majoring & Minorin

The Chemistry and Biochemistry Department offers majors and minors within the Bachelor of Science and Bachelor of Arts degrees.

The Chemistry and Biochemistry Department offers majors and minors within the BA and BS degrees. The department is accredited by the American Chemical Society (ACS) and offers an option for students to pursue an **ACS-certified BS degree in chemistry**. For students who wish to enter graduate school in chemistry or a related field upon completion of their degree, the ACS-certified chemistry major offers a strong foundation in all fields within the chemical sciences. The department also offers a major in biochemistry under the BS degree. The biochemistry curriculum is designed to guide students in developing a strong foundation in the fundamentals of chemistry and biology and is shaped by suggestions of the educational division of the American Society for Biochemistry and Molecular Biology (ASBMB).

In addition to their regular course work, chemistry and biochemistry majors are strongly encouraged to become involved in laboratory research during their junior and/or senior years. All chemistry and biochemistry majors are required to complete a capstone in their senior year.

NOTE: Students must earn a grade of C- or better for all prerequisite classes before enrolling in a given chemistry course.

Note on course numbering: 000 level courses are for non-majors. 100 level courses are introductory. 200-800 level courses are upper-level and require introductory or other courses as prerequisites. 900 level courses are capstone, research, tutorials, honors, internships, and independent study.

Major in Chemistry (BA): 15 courses

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- CHE51-201 Chemical Methods and Techniques Laboratory II
- CHE51-203 General Chemistry II
- CHE51-311 Organic Chemistry Laboratory Techniques
- CHE51-313 Organic Chemistry I ¹
- CHE51-321 Organic Chemistry Laboratory Applications
- CHE51-323 Organic Chemistry II ¹
- Four upper-level chemistry courses (300–700 level)
- Two additional advanced laboratory classes (800 level)
- CHE51-932 Senior Chemistry Capstone, or CHE51-922 Senior Biochemistry Capstone

Required supporting courses in the Chemistry major (BA): 2 courses

- MAT52-164 Modern Calculus I
- PHY53-154 Fundamentals of Physics I

¹ At least one course from CHE51-313 Organic Chemistry I and CHE51-323 Organic Chemistry II must be taken at Southwestern.

Major in Chemistry (BS): 17-18 courses

- CHE51-101 Chemical Methods and Techniques Laboratory I
- CHE51-103 General Chemistry I
- CHE51-201 Chemical Methods and Techniques Laboratory II
- CHE51-203 General Chemistry II
- CHE51-311 Organic Chemistry Laboratory Techniques
- CHE51-313 Organic Chemistry I ¹
- CHE51-321 Organic Chemistry Laboratory Applications

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- Three upper-level chemistry courses (400–700 level)
- Three additional advanced laboratory classes (800 level)
- Two credits of CHE51-91X Methods in Laboratory Research, **or** two credits of CHE51-94X Academic Internship, **or** one additional advanced laboratory course (800 level; this option is only available to students who have not completed a Methods in Laboratory Research course)
- CHE51-932 Senior Chemistry Capstone

Required supporting courses in the Chemistry major (BS): 6 courses

- MAT52-164 Modern Calculus I
- MAT52-264 Modern Calculus II
- BIO50-173/171 Molecular and Cellular Foundations of Biology /lab, **or** BIO50-183/181 Evolutionary and Ecological Foundations of Biology /lab
- PHY53-154 Fundamentals of Physics I
- PHY53-164 Fundamentals of Physics II
- One course from:
BIO50-173/171 Molecular and Cellular Foundations of Biology /lab
BIO50-183/181 Evolutionary and Ecological Foundations of Biology /lab
CSC54-184 Computer Science I
MAT52-364 Modern Calculus III
MAT52-674 Linear Algebra
PHY53-214 Modern Physics
PHY53-404 Electronics

¹ At least one course from CHE51-313 Organic Chemistry I and CHE51-323 Organic Chemistry II must be taken at Southwestern.

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CHE51-101 Chemical Methods and Techniques Laboratory I

- CHE51-103 General Chemistry I
- CHE51-201 Chemical Methods and Techniques Laboratory II
- CHE51-203 General Chemistry II
- CHE51-311 Organic Chemistry Laboratory Techniques
- CHE51-313 Organic Chemistry I ¹
- CHE51-321 Organic Chemistry Laboratory Applications
- CHE51-323 Organic Chemistry II ¹
- One class in Inorganic Chemistry (400 level)
- One class in Analytical Chemistry (500 level)
- One class in Biochemistry (600 level)
- CHE51-704 Physical Chemistry: Thermodynamics and Kinetics
- CHE51-714 Quantum Mechanics
- Two additional upper-level courses (4 credits each) approved by the Department Chair
- Three additional advanced laboratory classes (800 level)
- Two credits of CHE51-91X Methods in Laboratory Research
- CHE51-932 Senior Chemistry Capstone

Required supporting courses in the Chemistry major (BS): 6 courses

- MAT52-164 Modern Calculus I
- MAT52-264 Modern Calculus II
- BIO50-173/171 Molecular and Cellular Foundations of Biology /lab, or
BIO50-183/181 Evolutionary and Ecological Foundations of Biology /lab
- PHY53-154 Fundamentals of Physics I
- PHY53-164 Fundamentals of Physics II

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BIO50-163/161 Evolutionary and Ecological Foundations of Biology / lab

CSC54-184 Computer Science I

MAT52-364 Modern Calculus III

MAT52-674 Linear Algebra

PHY53-214 Modern Physics

PHY53-404 Electronics

¹ At least one course from CHE51-313 Organic Chemistry I and CHE51-323 Organic Chemistry II must be taken at Southwestern.

Major in Biochemistry (BS): 17-18 courses

- CHE51-101 Chemical Methods and Techniques Laboratory I
- CHE51-103 General Chemistry I
- CHE51-201 Chemical Methods and Techniques Laboratory II
- CHE51-203 General Chemistry II
- CHE51-311 Organic Chemistry Laboratory Techniques
- CHE51-313 Organic Chemistry I ¹
- CHE51-321 Organic Chemistry Laboratory Applications
- CHE51-323 Organic Chemistry II ¹
- CHE51-614 General Biochemistry I ²
- CHE51-624 General Biochemistry II ²
- One additional course in Biochemistry (600 level)
- One course from:

CHE51-404 Inorganic Chemistry and Biological Systems

CHE51-414 Organometallic Chemistry

CHE51-424 Fundamentals of Material Science and Engineering

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CHE51-704 Physical Chemistry: Thermodynamics and Kinetics

- One upper-level Biology course in the Cellular/Molecular area (300 level or above)
- CHE51-862 Advanced Lab in Biochemistry
- One additional advanced laboratory course (800 level)
- Two credits of CHE51-91X Methods in Laboratory Research, or two credits of CHE51-94X Academic Internship, or one additional advanced laboratory course (800 level; this option is only available to students who have not completed a Methods in Laboratory Research course)
- CHE51-932 Senior Chemistry Capstone, or CHE51-922 Senior Biochemistry Capstone

Required supporting courses in the Biochemistry major (BS): 6 courses

- MAT52-164 Modern Calculus I
- MAT52-264 Modern Calculus II, or MAT52-114 Introduction to Statistics
- BIO50-173/171 Molecular and Cellular Foundations of Biology/lab
- BIO50-183/181 Evolutionary and Ecological Foundations of Biology/lab
- PHY53-154 Fundamentals of Physics I
- PHY53-164 Fundamentals of Physics II

¹ At least one course from CHE51-313 Organic Chemistry I and CHE51-323 Organic Chemistry II must be taken at Southwestern.

² Both CHE51-614 General Biochemistry I and CHE51-624 General Biochemistry II courses must be taken at Southwestern

Minor in Chemistry: 11 courses

- CHE51-101 Chemical Methods and Techniques Laboratory I
- CHE51-103 General Chemistry I
- CHE51-201 Chemical Methods and Techniques Laboratory II

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- CHE51-313 Organic Chemistry I1
- CHE51-321 Organic Chemistry Laboratory Applications
- CHE51-323 Organic Chemistry II1
- Two upper-level chemistry courses (400-700) at different levels
- One advanced laboratory class (800 level)

See the Education Department for information regarding teacher certification in Chemistry.

GEORGETOWN, TEXAS

512.863.6511

CONTACT